

Classification Evaluation Metric -Decision Tree

Decision Tree classifier algorithm:

[76 9]

[8 41]

1. What is the percentage of correct classification of both (Purchase & Not purchased) to the total input of the test set?

Ans: Accuracy= 0.87

2. What is the percentage of correct classification of purchase to the total of purchase in the test set?

Ans: Recall of purchase = 0.84

3. What is the percentage of correct classification of not-purchase to the total of not purchase in the test set?

Ans: Recall of not purchase = 0.89

4. What is the percentage of correct classification of purchase to sum of correctly classified as purchase and wrongly classified as purchase in the test set?

Ans: Precision of purchase= 0.82

5. What is the percentage of correct classification of not purchase to sum of correctly classified as not purchase and wrongly classified as not purchase in the test set?

Ans: Precision of non purchase= 0.90

6. What is overall performance of purchase?

Ans: F1_measure of purchase = 0.83

7. What is overall performance of not purchase?

Ans: F1_measure of not purchase= 0.90

8. What is the average performance of precision (correctly and wrongly classified)?

Ans: macro_average of precision= 0.86

9. What is the average performance of recall (correctly classified)?

Ans: Macro_average of recall= 0.87

10. What is the average performance of F1_measure (overall performance)?

Ans: macro_average of F1_measure= 0.86

11. What is the sum of product of proportionate rate (weight class) of precision of each class?

Ans: Weight_average of precision= 0.87

12. What is the sum of product of proportionate rate (weight class) of recall of each class?

Ans: weight_average of recall=0.87

13. What is the sum of product of proportionate rate (weight class) of F1_measure of each class?

Ans: weight_average of F1_measure= 0.87